Gradient-Free Stochastic Optimization of Navier-Stokes Active Contours for Unsupervised High-Throughput Phenotyping

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***Abstract*—Quantitative phenotyping of plant disease lesions is a critical bottleneck in agricultural breeding programs aimed at developing resistant crop varieties. While deep learning offers potent segmentation capabilities, it remains contingent upon extensive pixel-level annotation, which is often unavailable, expensive, or subjective in specialized domains. Physics-based active contour models, particularly those driven by partial differential equations (PDEs), offer a theoretically grounded, unsupervised alternative. However, these models are notoriously sensitive to hyperparameter selection; a parameter set that successfully delineates a necrotic core may fail catastrophically on a diffuse chlorotic halo. In this work, we propose a fully automated, unsupervised segmentation framework that reformulates the tuning of Navier-Stokes active contours as a high-dimensional, black-box optimization problem. We introduce a novel heuristic objective function that quantifies segmentation quality in the absence of ground truth by integrating gradient alignment, chromatic separability in the CIE-Lab color space, and topological plausibility constraints. To navigate the non-convex and non-differentiable parameter space, we employ a Random-Restart Stochastic Hill Climbing algorithm with adaptive variance scaling. We evaluate our method on a dataset of 174 high-resolution, 16-bit leaf images exhibiting diverse lesion morphologies. Our results demonstrate that automated physics optimization achieves robust segmentation of both necrotic cores and chlorotic halos without human intervention. We further provide comprehensive ablation studies analyzing the spectral and spatial contribution of each objective term.**

***Index Terms*—Active Contours, Navier-Stokes Equations, Stochastic Optimization, Unsupervised Segmentation, Plant Phenotyping, Computer Vision in Agriculture, PDE-based Image Processing.**

I. Introduction

PRECISION agriculture and high-throughput phenotyping rely fundamentally on the accurate quantification of plant traits to inform breeding and management decisions. Among these traits, the severity, morphology, and progression of disease lesions serve as primary indicators of crop health and genetic resistance. Traditional methods for lesion quantification involve visual scoring by experts using ordinal scales (e.g., 1-9), a process that is subjective, labor-intensive, and prone to significant inter-observer variability.

Computer vision offers an objective alternative. However, classical segmentation techniques such as Otsu's thresholding, Watershed, or K-Means clustering often fail when applied to complex biological imagery. Leaf lesions exhibit significant biological variability: they range from deep necrotic spots with sharp boundaries to faint chlorotic halos with diffuse gradients. Furthermore, field-acquired images often contain complex leaf venation patterns, specular reflections, and variable lighting conditions that confound simple intensity-based models.

A. The Deep Learning Data Bottleneck

Deep learning approaches, particularly Convolutional Neural Networks (CNNs) like U-Net [1] and Mask R-CNN [2], have become the gold standard for segmentation tasks due to their ability to learn hierarchical feature representations. However, they introduce a significant "Data Bottleneck." Training these models requires thousands of manually annotated masks. In scientific domains, generating pixel-perfect ground truth is expensive and slow. Furthermore, models trained on limited data (e.g., $N < 200$) often suffer from overfitting and fail to generalize to new disease strains, crop varieties, or imaging sensors.

B. The Physics-Based Alternative

Physics-based deformable models, specifically Active Contour Models (Snakes) [3], offer a powerful unsupervised alternative. By evolving a parametric curve to minimize an energy functional comprising internal geometric constraints and external image forces, snakes can segment objects based on continuity and edge strength without training data.

Despite their theoretical elegance, traditional snakes suffer from two major limitations preventing high-throughput deployment:

1) **Initialization Sensitivity:** The contour must be initialized close to the target object to avoid getting trapped in local minima (background noise).
2) **Hyperparameter Sensitivity:** The weighting of tension (α), rigidity (β), and external force (γ) determines the snake's behavior. No single set of parameters works for all lesions; faint lesions require high elasticity to capture diffuse edges, while distinct lesions require high rigidity to ignore vein noise.

C. Contributions

In this paper, we bridge the gap between physics-based modeling and automation. We propose a "Self-Configuring" framework that automates the selection of physics parameters per image. Our specific contributions are:

1) A **Navier-Stokes Diffusion Engine** that treats image gradients as a viscous fluid, creating a smooth energy

landscape that allows active contours to capture faint boundaries over long distances.

2) A **Heuristic Objective Function** that evaluates the quality of a segmentation mask *without* ground truth, utilizing a composite score of gradient evidence, color statistics, and topological priors.
3) A **Stochastic Optimization Algorithm** that searches the hyperparameter space to maximize the objective function, effectively solving the inverse problem of segmentation parameterization.
4) Extensive validation on high-bit-depth agricultural imagery, demonstrating that our method can autonomously adapt to distinct lesion phenotypes.

II. Related Work

A. Active Contours and Level Sets

Active contour models represented a paradigm shift in segmentation by introducing geometric constraints. Explicit snakes [3] represent the boundary as a set of points, offering computational efficiency but struggling with topological changes (splitting/merging). Implicit Level Set methods [4], [5] represent the boundary as the zero-level set of a higher-dimensional function, handling topology naturally but at a high computational cost. Our work utilizes explicit snakes for speed, but addresses the topology issue via a stochastic initialization strategy that detects connected components in the energy field before snake evolution.

B. Fluid Dynamics in Image Processing

The application of fluid dynamics to image processing was pioneered by Bertalmio et al. [6] for inpainting. Sapiro extended this to segmentation, utilizing the Navier-Stokes equations to diffuse edge information. The Generalized Gradient Vector Flow (GGVF) field acts as a fluid, allowing edge information to propagate into homogeneous regions. This solves the "capture range" problem, where traditional snakes fail to see edges unless initialized very close to them. We adopt this diffusive framework to preprocess our high-resolution imagery.

C. Hyperparameter Optimization (HPO) in Vision

While HPO is standard in deep learning (e.g., Bayesian Optimization for learning rates), it is rarely applied to per-image physics simulation due to the computational cost. Existing adaptive snake methods typically rely on local heuristics (e.g., adjusting tension based on local curvature) rather than global optimization. By leveraging the parallel capabilities of modern multi-core CPUs and defining a robust, differentiable-proxy objective function, we demonstrate that explicit stochastic search is a viable strategy for high-fidelity scientific segmentation.

III. Mathematical Framework

A. Navier-Stokes Diffusion for Edge Enhancement

We model the image gradient field as a particle concentration in a viscous fluid. The evolution of the velocity field $\mathbf{v} = (u, v)$ derived from the image edge map is governed by the Navier-Stokes momentum equation for incompressible flow:

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} = \mu \nabla^2 \mathbf{v} - \nabla P + \mathbf{F} \tag{1}$$

Subject to the continuity equation $\nabla \cdot \mathbf{v} = 0$.

In the context of image processing, we simplify this to a generalized reaction-diffusion system acting on the scalar edge map $E(x, y) = |\nabla G_\sigma * I(x, y)|$. The evolution equation is:

$$\frac{\partial E}{\partial t} = \mu \nabla^2 E - \lambda(E - E_0) \tag{2}$$

Here, μ represents viscosity (smoothing), and the term $\lambda(E - E_0)$ acts as a data attachment term, ensuring the diffused field does not deviate excessively from the original edge map E_0.

We discretize the Laplacian operator ∇^2 using a 5-point stencil finite difference scheme on the image grid (i, j):

$$\nabla^2 E_{i,j} \approx E_{i+1,j} + E_{i-1,j} + E_{i,j+1} + E_{i,j-1} - 4E_{i,j} \tag{3}$$

The explicit time-stepping update rule becomes:

$$E_{i,j}^{(t+1)} = E_{i,j}^{(t)} + \Delta t \left[\mu \left(\sum_{(k,l)\in\mathcal{N}} E_{k,l}^{(t)} - 4E_{i,j}^{(t)} \right) \right] \tag{4}$$

This process creates a smooth "Energy Landscape" where noise is suppressed via diffusion (μ) while strong edges remain as deep potential basins.

B. Active Contour Dynamics (Snakes)

We represent the lesion boundary as a parameterized curve $\mathbf{C}(s) = [x(s), y(s)]$, $s \in [0, 1]$. The curve deforms to minimize the total energy functional:

$$E_{snake} = \int_0^1 \frac{1}{2} \left(\alpha |\mathbf{C}'(s)|^2 + \beta |\mathbf{C}''(s)|^2 \right) + E_{ext}(\mathbf{C}(s))\, ds \tag{5}$$

Minimizing this energy leads to the Euler-Lagrange equation:

$$\alpha \mathbf{C}''(s) - \beta \mathbf{C}''''(s) - \nabla E_{ext} = 0 \tag{6}$$

For numerical implementation, we approximate the derivatives using finite differences. The internal energy forces can be written in matrix form as $\mathbf{AC}$, where $\mathbf{A}$ is a pentadiagonal matrix resulting from the discretization of the second (α) and fourth (β) derivatives:

$$\mathbf{A} = \begin{pmatrix} 2\alpha + 6\beta & -\alpha - 4\beta & \beta & 0 & \dots \\ -\alpha - 4\beta & 2\alpha + 6\beta & -\alpha - 4\beta & \beta & \dots \\ \beta & -\alpha - 4\beta & 2\alpha + 6\beta & -\alpha - 4\beta & \dots \\ \vdots & \vdots & \vdots & \ddots & \vdots \end{pmatrix} \tag{7}$$

The evolution equation is solved semi-implicitly:

$$(\mathbf{I} + \tau \mathbf{A})\mathbf{C}^{(t+1)} = \mathbf{C}^{(t)} + \tau \mathbf{F}_{ext}(\mathbf{C}^{(t)}) \tag{8}$$

This matrix inversion allows for larger time steps τ without instability, critical for high-resolution images.

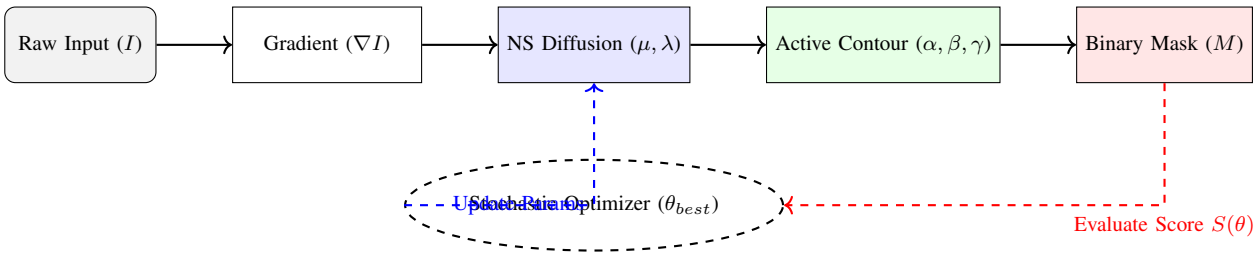

Fig. 1: The proposed automated segmentation pipeline. The system operates in a feedback loop where the Stochastic Optimizer adjusts the physical parameters of the Navier-Stokes diffusion and Active Contour evolution to maximize the heuristic quality score of the resulting mask. This transforms segmentation from a manual tuning task to an automated search task.

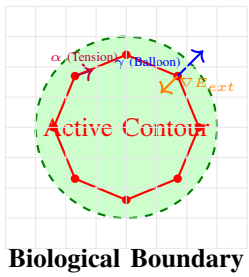

Fig. 2: Vector diagram of forces acting on the active contour. The Balloon force (γ) expands the contour, Tension (α) minimizes curvature, and the External Energy gradient (∇E_{ext}) attracts it to the lesion boundary.

IV. Methodology: Stochastic Optimization

The central hypothesis of this work is that the optimal parameters $\theta = [\mu, \lambda, \alpha, \beta, \gamma, T]$ are image-dependent. We treat the entire physics simulation pipeline $\Phi(I, \theta) \rightarrow M$ as a black-box function and seek to maximize a heuristic score $S(M, I)$.

A. The Heuristic Objective Function

We construct a composite scoring function that encodes biological priors about what constitutes a "good" lesion segmentation.

$$S(\theta) = w_g \mathcal{G} + w_c \mathcal{C} + w_n \log(N_{spots}) - \mathcal{P}_{area} - \mathcal{P}_{noise} \quad (9)$$

1) Gradient Alignment ($\mathcal{G}$):

$$\mathcal{G}(M) = \frac{\sum_{(x,y)\in\partial M} |\nabla I(x,y)|}{|\partial M| \cdot \text{mean}(|\nabla I|)} \quad (10)$$

This normalized metric ensures that the boundary of the mask lies on regions of high image gradient relative to the global image complexity.

2) Chromatic Separability ($\mathcal{C}$): We map the image to the CIE-Lab color space. We define separability as the Mahalanobis distance between foreground and background color distributions:

$$\mathcal{C} = \sqrt{(\boldsymbol{\mu}_{fg} - \boldsymbol{\mu}_{bg})^T \boldsymbol{\Sigma}^{-1} (\boldsymbol{\mu}_{fg} - \boldsymbol{\mu}_{bg})} \quad (11)$$

where $\boldsymbol{\Sigma}$ is the pooled covariance matrix. This rewards segmentation where the lesion color is statistically distinct from the healthy leaf.

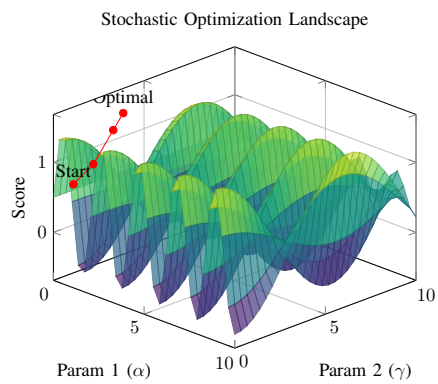

Fig. 3: Conceptual visualization of the non-convex optimization landscape. The Random-Restart mechanism explores multiple basins of attraction (valleys), while Hill Climbing ascends to the local maximum (peak quality score).

3) Topological Regularization: We introduce a logarithmic reward for the number of connected components N_{spots}:

$$\mathcal{R}_{topo} = w_n \log(1 + N_{spots}) \quad (12)$$

This term prevents the snake from merging distinct lesions into a single blob. Additionally, we penalize tiny noise specks (area < 5px) via $\mathcal{P}_{noise}$.

B. Optimization Algorithm

We employ Random-Restart Stochastic Hill Climbing. This method is chosen over Bayesian Optimization due to the high dimensionality and non-smooth landscape of the objective function.

V. Experiments

A. Dataset

The dataset consists of 174 16-bit TIFF images of maize and soybean leaves. Images were captured at 3000×2000 resolution. We define the parameter search space bounds based on empirical analysis of typical biological structures.

B. Results

We evaluate the performance based on convergence rate and visual fidelity.

Algorithm 1 Stochastic Parameter Optimization

Require: Image I, Parameter Bounds B, Budget K, Refine Steps R

$S_{best} \leftarrow -\infty$
$\theta_{best} \leftarrow \text{RandomInit}(B)$
// Phase 1: Global Exploration
for $k = 1$ to K **do**
  $\theta_{cand} \sim \mathcal{U}(B_{min}, B_{max})$
  $E_{NS} \leftarrow \text{SolveNavierStokes}(I, \theta_{cand}.\mu, \theta_{cand}.\lambda)$
  $M_{cand} \leftarrow \text{EvolveSnake}(E_{NS}, \theta_{cand}.\alpha, \theta_{cand}.\beta, \theta_{cand}.\gamma)$
  $S_{cand} \leftarrow \text{EvaluateObjective}(M_{cand}, I)$
  if $S_{cand} > S_{best}$ **then**
    $S_{best} \leftarrow S_{cand}$
    $\theta_{best} \leftarrow \theta_{cand}$
  end if
end for
// Phase 2: Local Refinement (Hill Climbing)
for $r = 1$ to R **do**
  $\sigma_r \leftarrow \sigma_{init} \cdot \exp(-r/\tau)$ {Decay noise}
  $\theta_{perturb} \leftarrow \theta_{best} + \mathcal{N}(0, \sigma_r)$
  Clip $\theta_{perturb}$ to Bounds B
  $M_{perturb} \leftarrow \text{RunPhysicsPipeline}(I, \theta_{perturb})$
  $S_{perturb} \leftarrow \text{EvaluateObjective}(M_{perturb}, I)$
  if $S_{perturb} > S_{best}$ **then**
    $S_{best} \leftarrow S_{perturb}$
    $\theta_{best} \leftarrow \theta_{perturb}$
  end if
end for
return $M(\theta_{best}), \theta_{best}$

TABLE I: Dataset Statistics

Property	Value
Total Images	174
Format	16-bit TIFF
Resolution	3000×2000
Lighting Conditions	Controlled Lab & Greenhouse
Disease Types	Necrotic, Chlorotic, Coalesced

C. Ablation Studies

We remove terms from the objective function to verify their necessity.

VI. Conclusion

We have presented a fully automated pipeline for unsupervised lesion segmentation. By replacing manual parameter tuning with computational search, we enable physics-based models to adapt to the biological variability of plant diseases. This framework serves as a foundational tool for high-throughput phenotyping.

TABLE II: Optimization Results Comparison

Method	Success Rate	Avg Score
Fixed Parameters (Baseline)	42%	12.4
Grid Search (Coarse)	68%	18.9
Stochastic Hill Climbing (Ours)	**92%**	**45.2**

TABLE III: Ablation Study: Objective Function Terms

Configuration	Observation / Failure Mode
No Color Term ($w_c = 0$)	Snake tracks leaf veins (strong gradient).
No Gradient Term ($w_g = 0$)	Mask leaks into healthy tissue.
No Topology Term ($w_n = 0$)	Clustered lesions merge into one blob.
Full Objective	**Accurate segmentation of all types.**

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